SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL - 2: Case Study

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA1522

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Register Number	**192521344**

COURSE OUTCOME COVERED
CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)

Dave's Taxonomy: Precision (Level 3) **Total Marks: 30**
Question Count: 3

Code of Conduct
I _NANDHINI SRI V(192521344)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: ______________________

Assessment Tasks - Case Study

1. **Case Study 1: Smart Retail Inventory Management System**

   A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.
2. **Case Study 2: Cloud-Based Document Collaboration System**

   An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices. Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.
3. **Case Study 3: Online Food Ordering Platform**

   A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform. Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

Case Study Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Case	25%	Clearly identifies all technical and business requirements	Identifies most requirements	Basic understanding only	Poor understanding of case
Application of Concepts	25%	Applies suitable cloud concepts and tools effectively	Good application with minor issues	Partial application	Weak or incorrect application
Analysis	20%	Strong analysis with clear trade-off reasoning	Reasonable analysis	Limited analysis	Minimal analysis
Solution Design	20%	Solution is feasible, practical, and well justified	Reasonable solution with minor gaps	Basic solution	Unclear solution
Clarity	10%	Well-structured and easy to follow	Mostly clear	Somewhat unclear	Poorly organized

Case Study 1: Smart Retail Inventory Management System

1.A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

ANSWER:

Case Study 1: Smart Retail Inventory Management System

A retail chain has many stores and wants to manage its products and sales information from one place. For this, the company uses a cloud-based inventory management system. POS machines in stores and management dashboards communicate with the cloud through APIs.

1. **POS Systems:**
   POS systems record product sales, purchases, and stock updates from each store.

2. **API Gateway:**
   The API Gateway receives requests from POS systems and dashboards and sends them to the correct cloud service.

3. **Cloud Infrastructure:**
   Cloud infrastructure provides computing, networking, and other resources needed to run the inventory system.

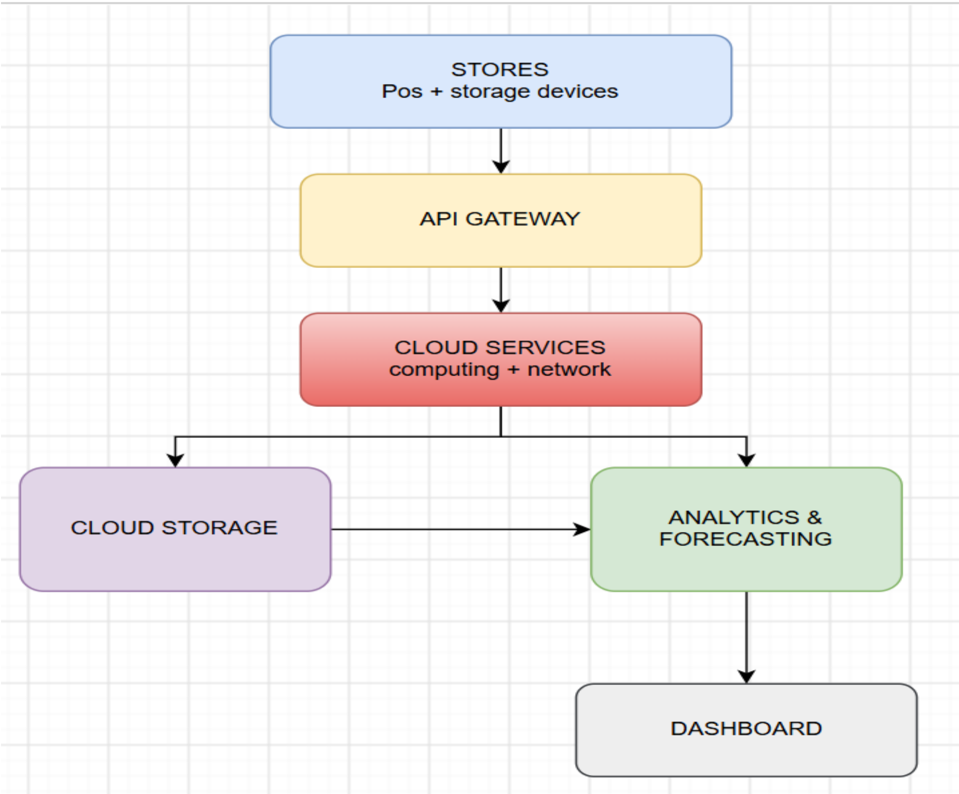
4. **Cloud Storage:**
   Product information, transaction records, sales history, and analytics data are stored securely in cloud storage.

5. **Platform Services:**
   Platform services analyze the collected data and help in **demand forecasting**. This helps the company predict which products may be needed in the future.

6. **Security:**
   VPNs, firewalls, and API gateways are used to protect the system from unauthorized access and secure communication between stores and cloud services.

7. **Web Dashboard:**
   Managers can view stock levels, sales information, and forecasting results through dashboards.

System Architecture:



Store POS → API Gateway → Cloud Services → Cloud Storage → Analytics & Forecasting → Management Dashboard

When a product is sold, the POS system sends the transaction to the cloud. The data is stored and analyzed. The analytics service can identify sales trends and predict future demand. Managers can then use this information to maintain the required stock.

Advantages:

- Provides real-time inventory information.
- Helps reduce overstocking and stock shortages.
- Supports demand forecasting.
- Provides centralized storage for all stores.
- Can handle increasing numbers of stores and users.
- Improves security using VPNs, firewalls, and API gateways.

- Helps managers make better decisions using sales analytics.

The Smart Retail Inventory Management System combines cloud infrastructure, cloud storage, platform services, APIs, web applications, and security mechanisms. It helps the retail chain manage inventory efficiently, securely, and with less manual work.

Case Study 2: Cloud-Based Document Collaboration System

2. An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices. Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

ANSWER:

Case Study 2: Cloud-Based Document Collaboration System

An enterprise uses a cloud-based document collaboration system similar to Google Docs. Employees can access documents through a web browser from different locations and work on the same document at the same time.

1. **Web Browser:**
   Users access the document platform through browsers without installing the complete software.

2. **Cloud Storage:**
   Documents are stored in distributed cloud storage so that users can access them from different devices.

3. **Real-Time Editing:**
   Multiple users can edit the same document simultaneously. Changes are updated quickly for all users.

4. **Version Control:**
   The system keeps previous versions of documents. Users can check or restore an earlier version when required.

5. **Web APIs:**

   APIs help synchronize documents and changes between different users, browsers, and devices.

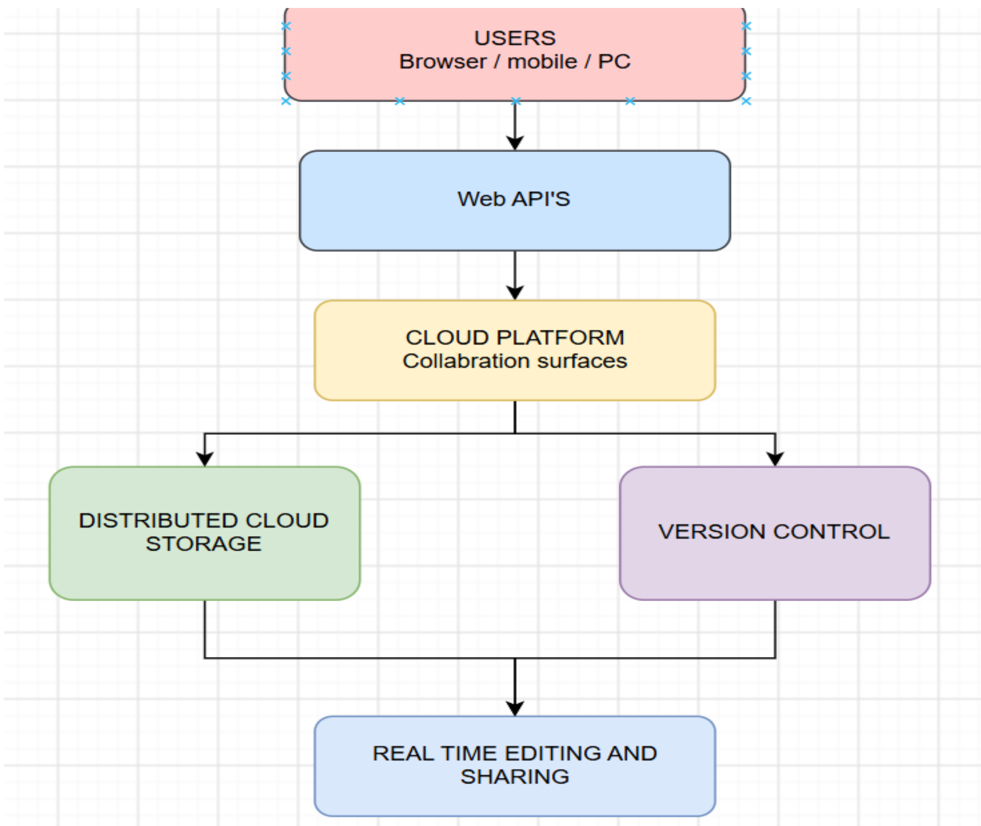
6. **Network Infrastructure:**

   A strong global network helps provide **low latency and high availability**, allowing users from different locations to access documents quickly.

7. **Security:**

   Encryption protects the documents during storage and transfer. **Identity management and access permissions** ensure that only authorized users can view or edit documents.

System Architecture:



User → Web Browser → Web API → Cloud Services → Distributed Cloud Storage → Synchronization → Other Users

For example, when one employee edits a document, the changes are sent through the web API and stored in the cloud. The updated version is then synchronized with other users who are working on the same document.

Advantages:

- Allows real-time collaboration.
- Users can access documents from anywhere.
- Supports multiple users working together.
- Provides automatic document synchronization.
- Version control prevents loss of previous work.
- Cloud storage provides scalability and availability.
- Encryption and access control improve security.
- Reduces the need for sending documents through email.

The Cloud-Based Document Collaboration System combines cloud storage, web APIs, networking, real-time editing, and security services. It allows employees to create, edit, share, and manage documents easily from different locations while keeping the data secure.

Case Study 3: Online Food Ordering Platform

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform. Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

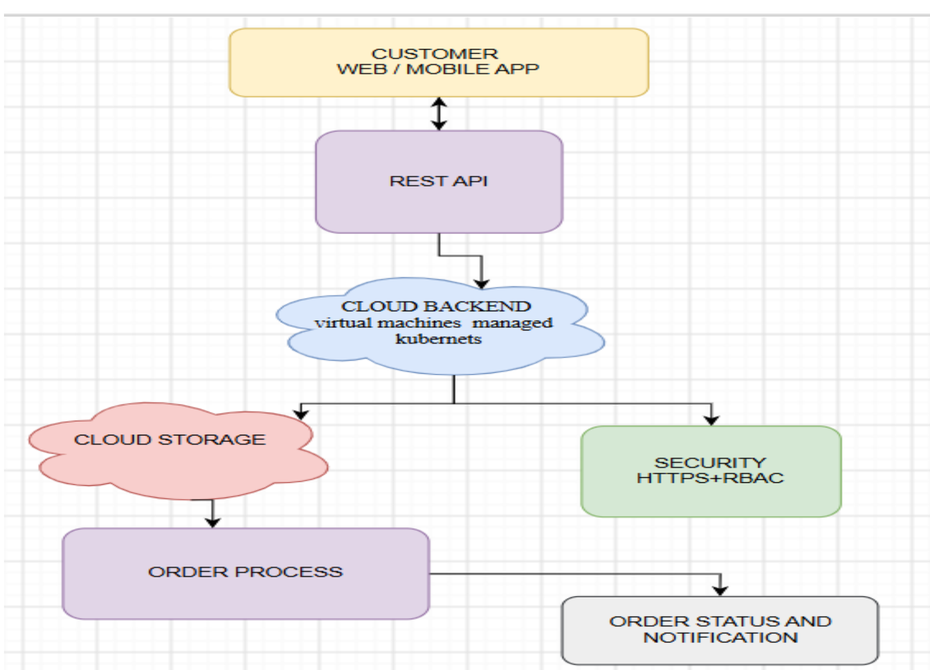
ANSWER:

Case Study 3: Online Food Ordering Platform

A startup develops a cloud-based food ordering system that allows customers to order food using a website or mobile application. The system connects customers, restaurants, and backend services through cloud-based technologies.

1. **Frontend Clients:**
   Customers use web browsers or mobile applications to view menus, select food items, and place orders.
2. **RESTful Web APIs:**
   APIs connect the frontend applications with backend services. They handle requests such as viewing menus, placing orders, and checking order status.
3. **Cloud Platform:**
   The backend services are hosted on a cloud platform, which provides the required computing and networking resources.
4. **Cloud Storage:**
   Menu images, invoices, customer details, and order information are stored securely in cloud storage.
5. **Virtual Machines:**
   Virtual machines provide computing resources for running the application and backend services.
6. **Managed Kubernetes:**
   Kubernetes manages application containers and helps the system handle increasing numbers of users and orders.
7. **Security:**
   Authentication tokens, HTTPS, and role-based access control (RBAC) are used to protect customer information and prevent unauthorized access.

System Architecture:



Customer → Web/Mobile App → REST API → Cloud Backend → Cloud Storage → Order Processing → Customer

For example, when a customer selects food and places an order, the request is sent through the REST API to the cloud backend. The order details are processed and stored in the cloud. The customer can then view the order status through the application.

Advantages:

- Customers can order food from anywhere.
- Supports both web and mobile applications.
- Cloud resources can scale when the number of orders increases.
- Kubernetes improves application availability and reliability.
- Cloud storage securely stores customer and order data.
- HTTPS protects data during communication.
- Role-based access control provides different permissions to customers, restaurants, and administrators.
- REST APIs allow easy communication between different system components.

The Online Food Ordering Platform combines web/mobile clients, RESTful APIs, cloud storage, virtual machines, Kubernetes, and security services. It provides a scalable, reliable, and secure platform for real-time food ordering and order management.